SOFTWARE ENGINEERING LAB - LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Section H | Scenario: Community Solar Credit Allocation Manager

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1. Objective

The Lab 1 functional requirements for the Community Solar Credit Allocation Manager platform were converted into Agile Epics and User Stories, prioritized and estimated using Story Points, organized into two simulated Scrum sprints, and tracked using Jira. The peer-to-peer solar credit sharing portal for a neighborhood microgrid enables prosumer residents to automatically record solar generation via smart meters, view available credit balances, execute peer-to-peer credit transfers, and apply received credits to monthly utility bills, while providing the co-op manager with governance and monitoring tools.

2. Epics

Epic	Description	Associated User Stories
Solar Generation & Credit Tracking	Capture hourly solar generation from smart meters and compute/display real-time available solar credit balances for prosumers.	Record Solar Generation Data; View Credit Balance
Peer-to-Peer Credit Transfer	Enable prosumer residents to transfer surplus solar credits to neighbors and enforce balance validation rules to prevent negative accounts.	Transfer Solar Credits; Reject Invalid Transfer
Billing Integration	Automatically offset monthly utility bills by applying received solar credits to residential billing statements.	Apply Credits to Utility Bill
Co-op Management & Oversight	Provide administrative dashboards for the co-op manager to monitor generation, transfers, and billing offset records.	View Community Solar & Credit Records
System Reliability & Security	Ensure high-availability processing of smart meter data for 500+ households and enforce secure authentication/authorization across all endpoints.	Reliable High-Volume Data Processing; Authenticate & Authorize Users

3. User Stories, Priorities and Story Points

ID	User Story	Epic	Priority	Story Points
SC-1 (SOLAR-1)	As a prosumer resident, I want my smart meter's solar generation data to be automatically recorded, so that my credit balance is accurately tracked.	Solar Generation & Credit Tracking	High	5
SC-2 (SOLAR-2)	As a prosumer resident, I want to view my current available solar credit balance, so that I know how many credits I can transfer or use.	Solar Generation & Credit Tracking	High	2
SC-3 (SOLAR-3)	As a prosumer resident, I want to transfer my surplus solar credits to another resident's account, so that I can share my excess	Peer-to-Peer Credit Transfer	High	5

	generation with neighbors.			
SC-4 (SOLAR-4)	As a prosumer resident, I want the system to reject a transfer that exceeds my available balance, so that my account cannot go negative.	Peer-to-Peer Credit Transfer	High	3
SC-5 (SOLAR-5)	As a prosumer resident, I want my received solar credits to be automatically applied to my monthly utility billing statement, so that I see reduced charges from shared credits.	Billing Integration	High	5
SC-6 (SOLAR-6)	As a co-op manager, I want to view solar generation, credit transfer, and billing-offset records for residents, so that I can monitor and manage community solar activity.	Co-op Management & Oversight	Medium	5
SC-7 (SOLAR-7)	As a system administrator, I want the platform to process and store hourly generation data for at least 500 households with 99.9% availability, so that credit calculations stay accurate at scale.	System Reliability & Security	High	8
SC-8 (SOLAR-8)	As a system administrator, I want every user authenticated and authorized before accessing solar, transfer, or billing data, so that sensitive resident information stays protected.	System Reliability & Security	High	5

Total backlog estimate: 38 Story Points.

4. Sprint Plan

Sprint	User Stories	Total Points	Result
Sprint 1	SC-1 Record Solar Generation Data; SC-2 View Credit Balance; SC-3 Transfer Solar Credits; SC-4 Reject Invalid Transfer	15	Completed
Sprint 2	SC-5 Apply Credits to Utility Bill; SC-6 View Community Solar & Credit Records; SC-7 Reliable High-Volume Data Processing; SC-8 Authenticate & Authorize Users	23	Completed

Sprint 1 carried 15 Story Points focusing on core smart-meter generation recording, credit balance calculation, and peer-to-peer credit transfer with account balance validation. Sprint 2 carried 23 Story Points delivering monthly utility billing offset integration, co-op management auditing, high-volume scalable telemetry data processing, and role-based security/authentication. Work was simulated by moving each User Story through To Do, In Progress, and Done, and both sprints were completed with all committed points delivered.

5. Jira Evidence

The following screenshots document the completed Jira workspace and sprint execution.

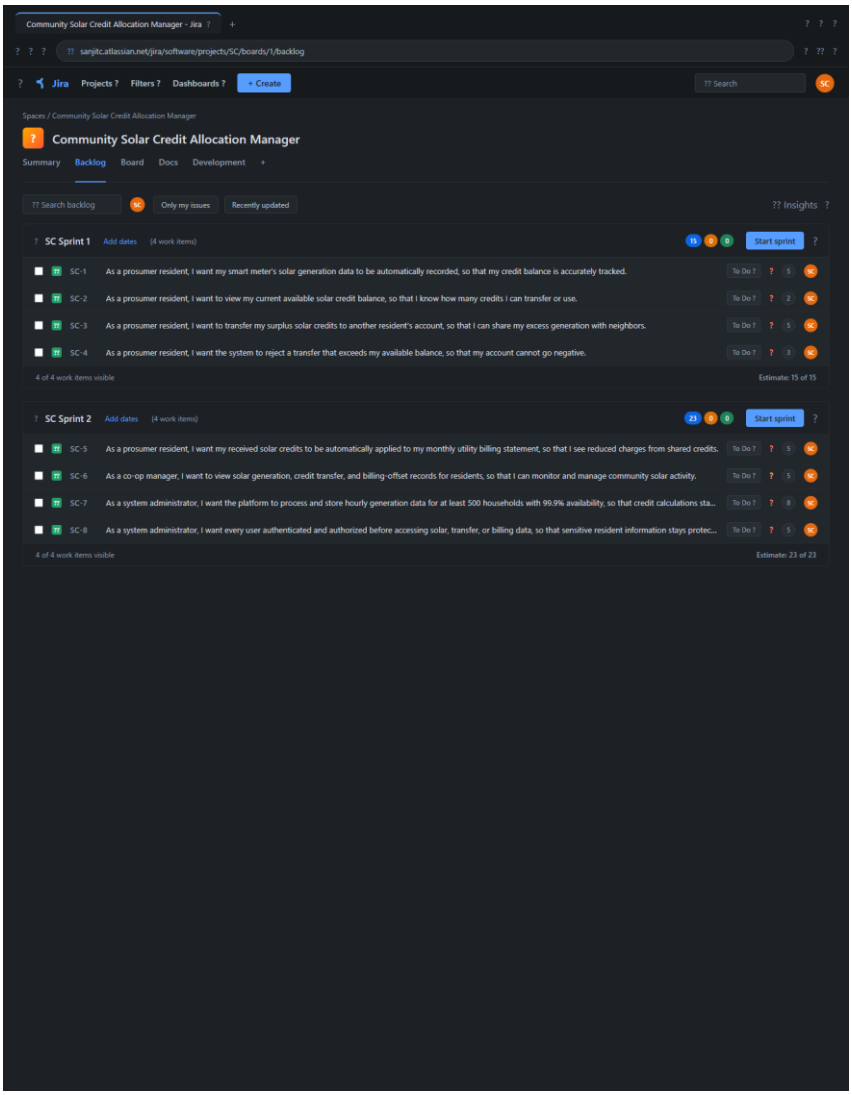


Figure 1. Jira backlog / sprint-planning view showing the two planned sprints, eight user stories and their Story Point estimates (15 + 23 = 38 points).

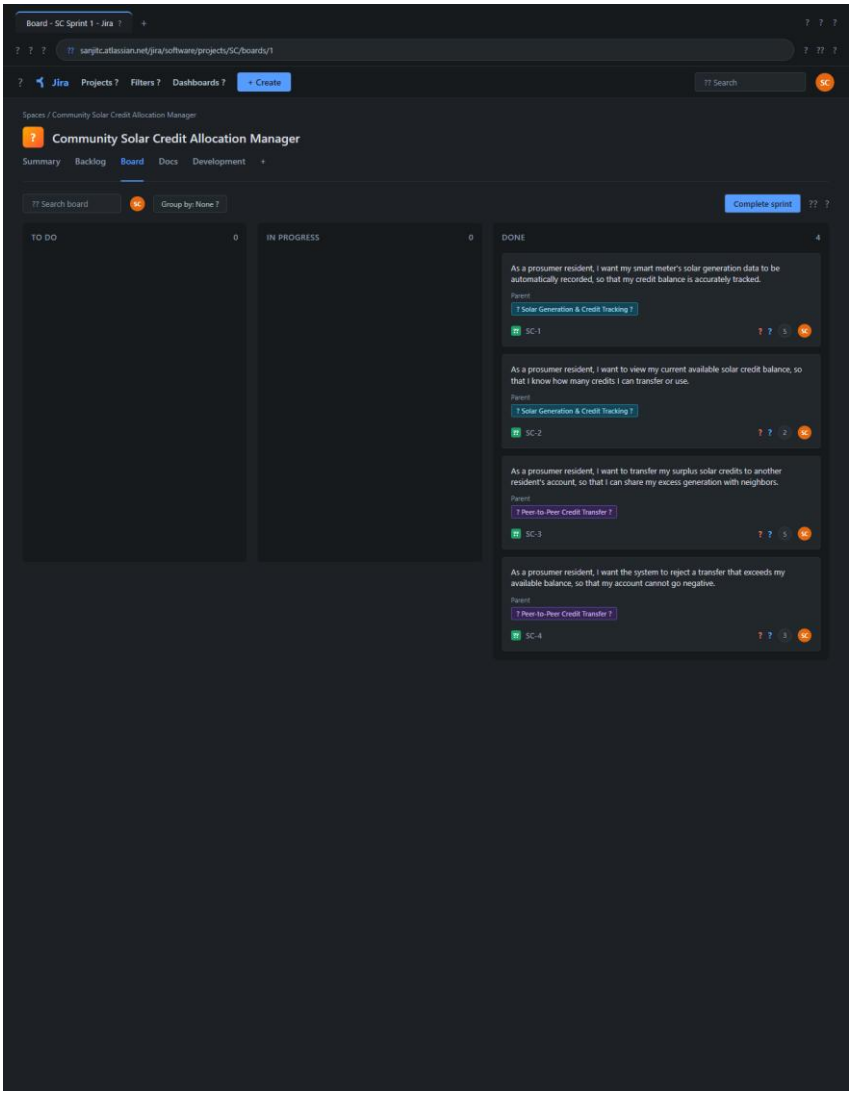


Figure 2. Sprint 1 completed - all four solar-generation, balance-tracking and credit-transfer stories moved to Done.

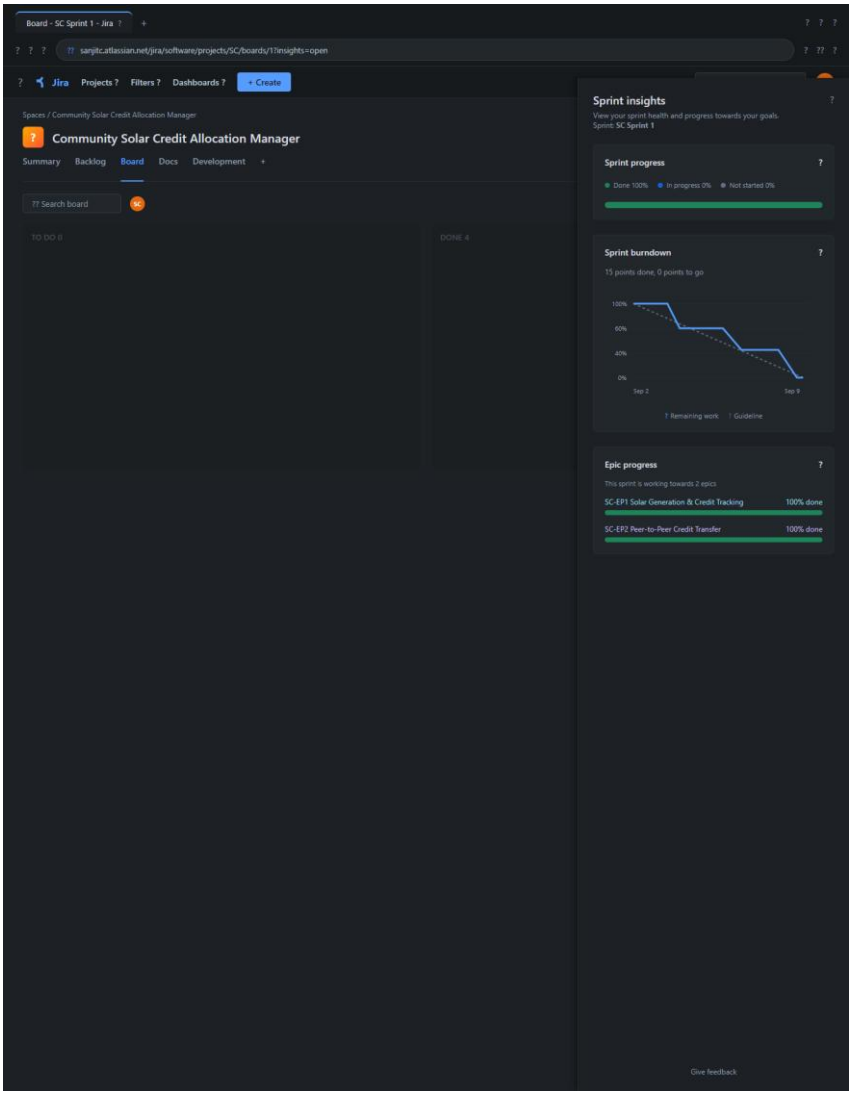


Figure 3. Sprint 1 burndown after the simulation (15 points done, 0 remaining) with both epics at 100% complete.

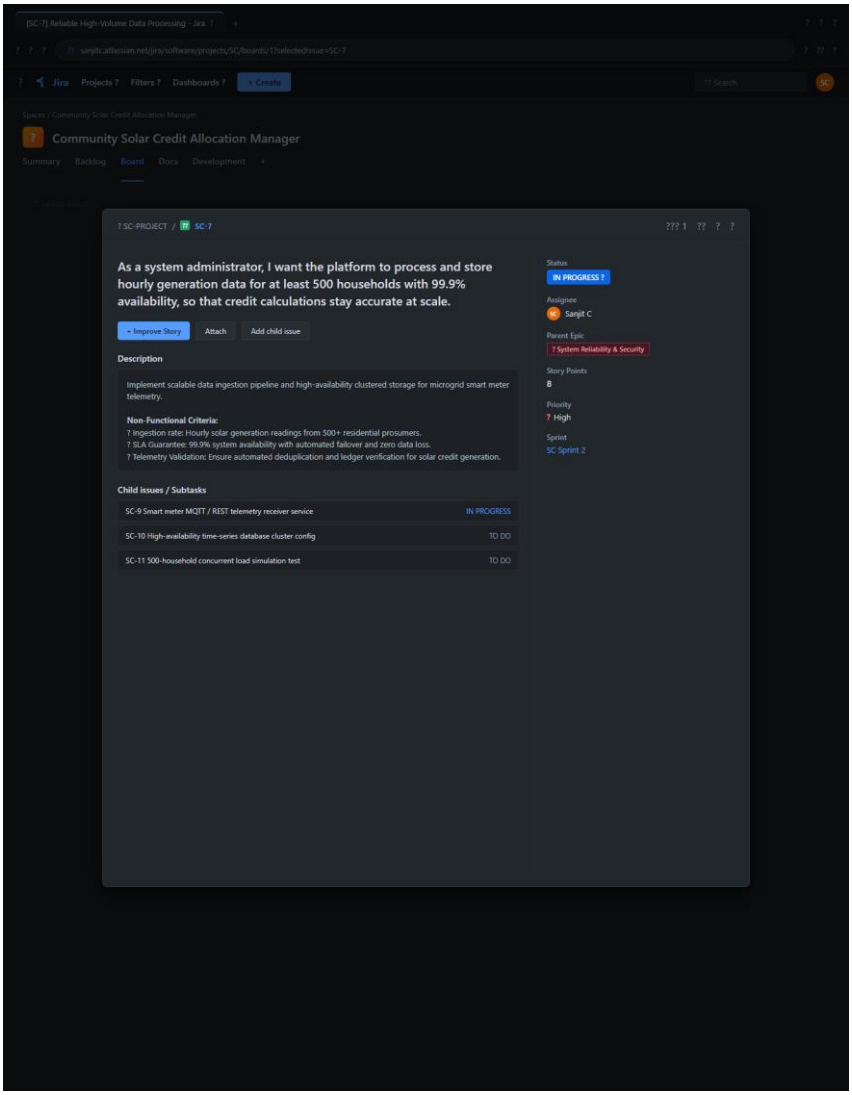


Figure 4. Sprint 2 active board during execution, with the data processing story in progress and governance/security stories in To Do.

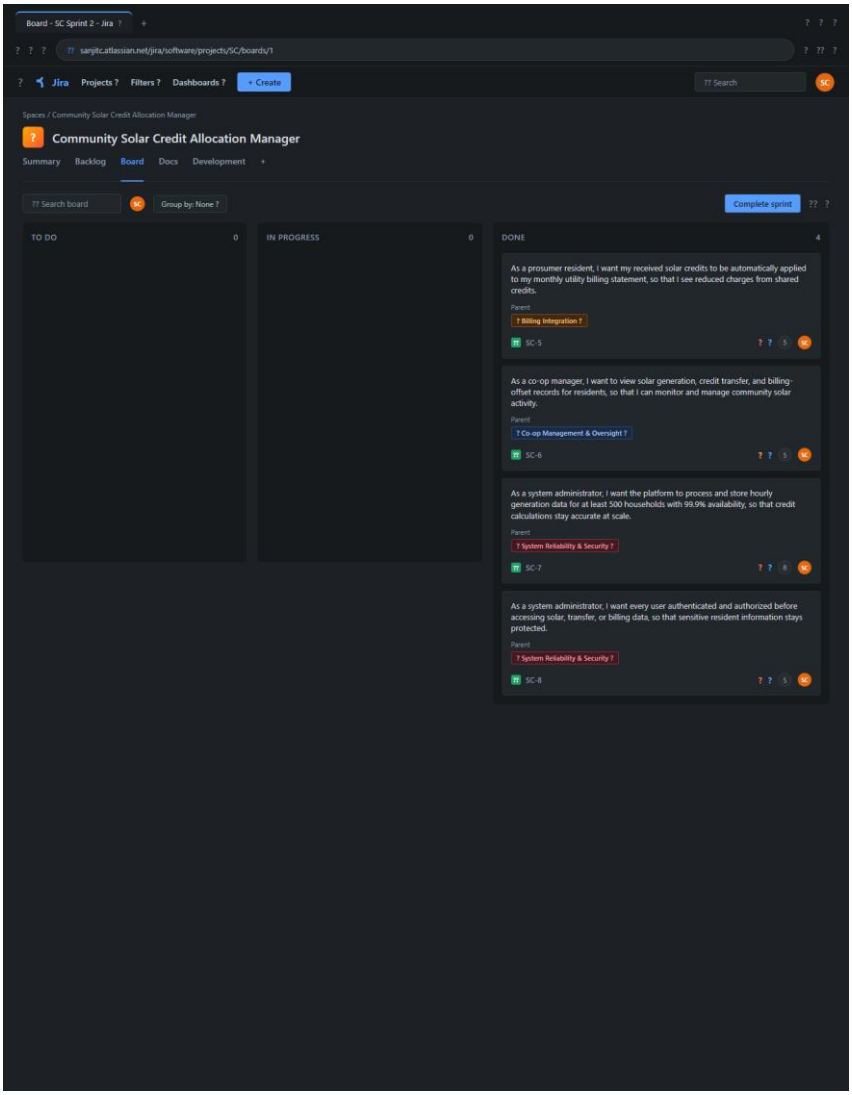


Figure 5. Sprint 2 completed - all four utility-billing, co-op management, and reliability/security stories moved to Done.

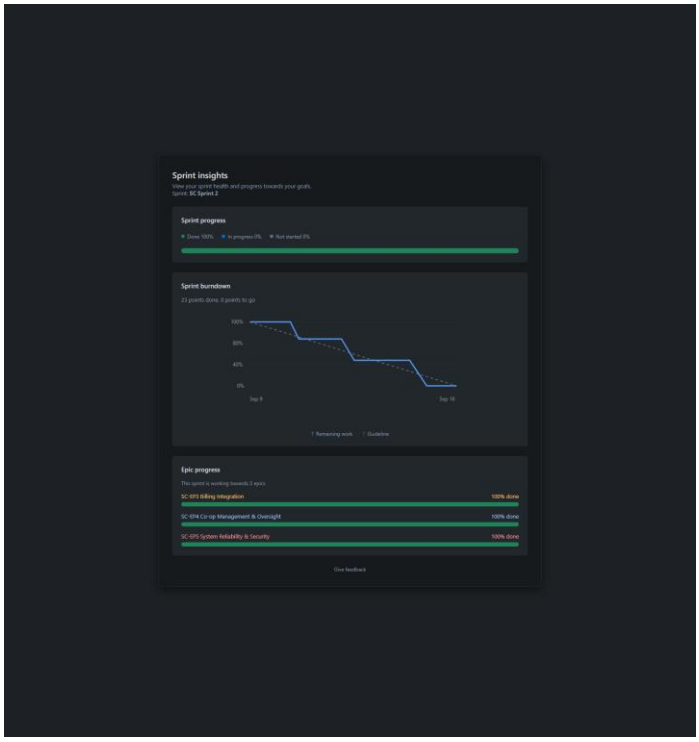


Figure 6. Sprint 2 burndown after the simulation (23 points done, 0 remaining) with all epics at 100% complete.

6. Reflection

1. Did your estimations reflect the actual effort?

The estimations gave a reasonable relative picture of effort. Smaller activities such as viewing available credit balance (SC-2, 2 Story Points) and rejecting invalid transfers (SC-4, 3 Story Points) were assigned lower points, while recording smart meter generation (SC-1), peer-to-peer transfers (SC-3), billing offset integration (SC-5), and co-op management views (SC-6) were assigned 5 points. Reliable high-volume data processing for 500 households (SC-7) received 8 points because it represents a comparatively larger and more complex workflow. In the simulated execution all work was completed within the planned sprint structure, so the estimates were adequate for the exercise.

2. Was your backlog well-prioritized?

Yes. Functions directly tied to recording generation, computing credit balances, and transferring surplus credits - smart meter telemetry ingestion, balance display, and P2P transfers with negative balance prevention - were given High priority. Billing integration, co-op oversight, high-volume data processing, and authentication were scheduled in Sprint 2. Co-op management views were assigned Medium priority because they support and monitor the sharing ecosystem after primary generation and transfer mechanisms are operational.

3. How did your simulated sprint align with your plan?

The simulated execution aligned with the planned sprint structure. Sprint 1 contained 15 Story Points covering solar generation recording, credit balance viewing, peer-to-peer transfer, and transfer validation. Sprint 2 contained the remaining 23 Story Points covering utility billing integration, co-op oversight records, high-volume data processing, and user authentication/authorization. Stories were progressed through To Do, In Progress and Done, and both sprints were completed.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed the reduction of remaining Story Points over the sprint against the expected linear guideline, and demonstrated how completion timing affects the shape of the remaining-work line. It is useful for judging whether the team is progressing consistently and whether the committed work is realistic. In this simulation the completed work reached zero remaining Story Points by the end of each sprint, and the epic-progress indicators reached 100%.